Haseeb Ashfaq
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EXPERIENCE

Google **May 2025 – August 2025**
Software Engineering Intern, Ph.D. California, USA

- Part of the AI and Infrastructure organization, worked on improving the robustness of ML infrastructure
- Developed a tool, *hstprof*, for profiling GPU/TPU workloads using high-speed network telemetry
- *hstprof* enabled a fine-grained view of the network traffic of ML jobs, reporting utilization at a 100-microsecond timescale
- Used *hstprof* to analyze Gemini training clusters to investigate several regressions, including packet drops and high latencies
- Found an imbalance of packet queues across the memory banks of switches, which caused performance degradation

Nokia Bell Labs **June 2023 – August 2023**
Networking Research Intern New Jersey, USA

- Developed (in C++, Unix) a streaming service for AR/VR content for heterogeneous networks
- Implemented a resource-efficient transcoding mechanism for volumetric videos that achieved 75% CPU savings
- Developed an encoder/decoder for point cloud data that can tolerate packet losses in the network, which enabled utilizing an unreliable transport protocol (UDP) instead of TCP for point cloud streaming

Networks and Distributed Systems Group, NYU **Sep. 2025 – May 2026**
Graduate Research Assistant New York, USA

- Designed a special priority queue, LOQ, for cloud-hosted financial exchanges, that enhances a matching engine's throughput by up to 150% and lowers latency by 90% while retaining fairness among market participants
- Designed a cloud multicast service, in C++, for market data, delivering lower latency and greater scalability than AWS

PosterMyWall **June 2020 – August 2021**
Software Engineer (Full Time) Lahore, PK

- Developed, in PHP and JS, an access control system for internal tools of the company
- Set up a CI/CD pipeline along with testing infrastructure using TeamCity and AWS
- Automated AWS-hosted development infrastructure, shortening the testing cycle time by more than 50%
- Secured the product website by eliminating critical vulnerabilities (XSS, CSRF, IDOR) and did backend development

EDUCATION

PhD and MS, Computer Science **Sep. 2021 – May 2026**
New York University, New York, USA GPA: 4.0/4.0
Research Interests: Distributed Systems, Networks, Cloud Computing, Financial Technologies, AI Infrastructure

Bachelor of Science, Computer Science **Sep. 2016 – May 2020**
Lahore University of Management Sciences, Lahore, Pakistan GPA: 3.7/4.0
Courses: Algorithms, Data Structures, Distributed Systems, Computer Networks, Machine Learning

PROJECTS

Design and Implementation of a Low-Latency and Scalable Cloud Financial Exchange

- Implemented a low-latency market data multicast, achieving a 1-microsecond latency difference across receivers
- Utilized DPDK kernel bypass and zero-copy packet replication techniques, implemented in C++
- Utilized eBPF/XDP and eBPF/TC for efficient packet processing when using the Linux kernel

Codesign Of Tensors' Encoding And Transcoding For Decentralized ML

- Designed and implemented a mechanism for packing tensors in network packets that enables a resource-efficient transcoding (creating various bitrate versions) mechanism, akin to Scalable Video Codec, but for tensors
- Enabled utilizing overlay multicast for distributing training data across geo-distributed heterogeneous clients
- Reduced memory utilization by 30% and increased the throughput of data dissemination by 25%

KV Cache-Optimized Enterprise LLM Coding Agent

- Designed a coding agent that maximized KV cache usage across members of the same team, reducing inference cost
- As several engineers of a team share base source code, the coding agent constructs prompts as a combination of base source files and per-engineer patches, along with several optimizations to enhance the common prefix across prompts
- Reduced LLM prefill computation substantially through ~85% resultant common prompt prefixes

PUBLICATIONS

Beyond Lamport, Towards Probabilistic Fair Ordering
ACM HotNets'25

Network Support For Scalable and High-Performance Cloud Exchanges
ACM SIGCOMM '25, Cited by **Jane Street**, **Google,** and **Microsoft Research**

ParserHawk: Hardware-aware parser generator using program synthesis
ACM SIGCOMM '25

Towards Efficient Transmission of 3D Point Clouds Via Adaptive Encoding and QUIC
Emerging Multimedia Systems (EMS)'25, a SIGCOMM workshop

Codesign of Tensors Encoding And Transcoding
Networks for AI Computing (NAIC)'25, a SIGCOMM workshop

Patent: A Method To Enable Fast Transmission And Processing Of 3D Telepresence Data Encoded As Octrees
Approved by Nokia's internal board, in submission to USPTO, received a monetary award from Nokia Bell Labs

QuEST: Fast, Expressive, and Cheap Analytics for Distributed Traces Using Cloud Storage
CloudDB, a VLDB workshop

INVITED TALKS

HotNets'25: Probabilistic Fair Ordering of Events. 11/18/2025
Open Compute Project, OCPTAP: Network Support For Scalable And High Performance Cloud Exchanges. 10/22/2025
Google: How to build an ultra-fast and scalable financial exchange on the public cloud? 12/03/2024
Rutgers University: Network support for cloud-hosted financial exchanges. 10/30/2024

AWARDS, FELLOWSHIPS, AND SERVICES

Outstanding Research Award
Granted by Nokia Bell Labs during the Global Student Program 2023

National Science Foundation (NSF) Grant
Funds for traveling to ACM Sigcomm 2024 in Sydney, Australia

Reviewer for academic research articles
Served as a reviewer for research articles submitted to ACM JCSS and IEEE Transactions on Cloud Computing

MacCracken Fellowship
Granted by New York University for a Ph.D. in Computer Science

SKILLS

C/C++, Python, PHP, SQL, Go, JavaScript, React/React Native, Rust, AWS, Debugging, Testing, DPDK, eBPF, Linux, Kubernetes, Docker, Istio, Microservices, Congestion Signalling (CSig), High Speed Network Telemetry, System Design, KV Cache Optimization

MISC.

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Available for full-time: June 2026